Discrete Structure SECI1013 - section 08

Subject: Assignment 2

No:

Date:

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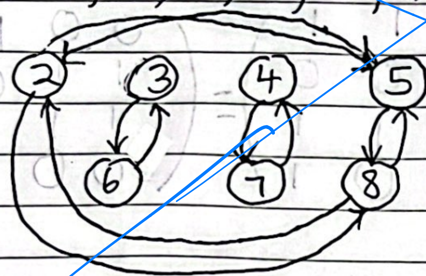
Question 1

R = { (2,5), (5,2), (3,6), (6,3), (4,7), (7,4), (2,8), (8,2), (5,8), (8,5) }

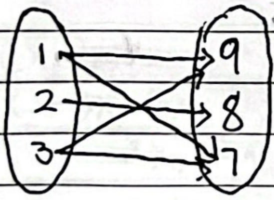


(2)(a) A = {1, 2, 3} B = {9, 8, 7}

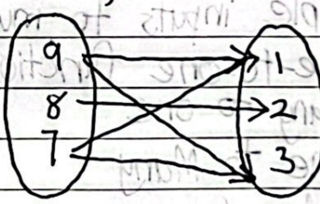
R = {(1,9), (1,7), (2,8), (3,9), (3,7)}

R^{-1} = {(9,1), (7,1), (8,2), (9,3), (7,3)}

(b)

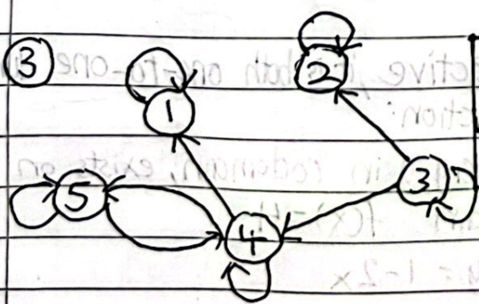


R^{-1}



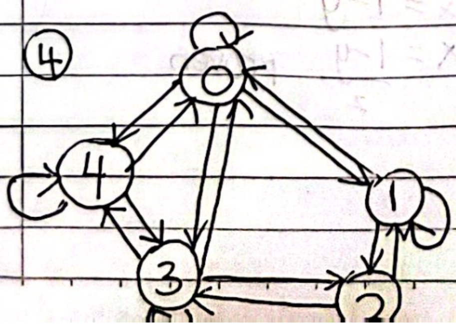
(c) The inverse relation R^{-1} is formed by swepping the element in each orderpair of original relation, R. R^{-1} can be described as For all $(b,a) \in B \times A$, and given $bR^{-1}a \Leftrightarrow b+a$ is even

(3)



	1	2	3	4	5
in degree	2	2	1	3	2
out degree	1	1	3	3	2

(4)



$$\begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{array}\begin{array}{c} \begin{array}{ccccc} 0 & 1 & 2 & 3 & 4 \end{array} \\ \begin{pmatrix} 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \end{pmatrix} \end{array}$$

symmetric, reflexive, not transitive

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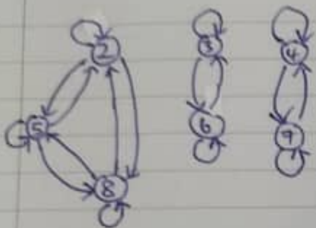
No: Date:

$R = \{(2,5),(5,2),(3,6),(6,3),(4,7),(7,4),(2,8),(8,2),(5,8),(8,5),$
$(2,2),(3,3),(4,4),(5,5),(6,6),(7,7),(8,8)\}$



R is reflexive relation since the diagraph has a loop at every vertex for each element $x \in A$, $(x,x) \in R$.

R is symmetric relation since whenever it is a directed edge from x to y, there is also a directed edge from y to x.

Let M_R be matrix of relation R

Let $M_R \otimes M_R = N$

Let $M_R = [m_{ij}]$, $N = [n_{ij}]$

$$\begin{array}{c c}
 & \begin{array}{ccccccc} 2 & 3 & 4 & 5 & 6 & 7 & 8 \end{array} \\
\begin{array}{c} 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \end{array} &
\begin{bmatrix}
1 & 0 & 0 & 1 & 0 & 0 & 1 \\
0 & 1 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 1 & 0 \\
1 & 0 & 0 & 1 & 0 & 0 & 1 \\
0 & 1 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 1 & 0 \\
1 & 0 & 0 & 1 & 0 & 0 & 1
\end{bmatrix}
\end{array}
\otimes
\begin{bmatrix}
1 & 0 & 0 & 1 & 0 & 0 & 1 \\
0 & 1 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 1 & 0 \\
1 & 0 & 0 & 1 & 0 & 0 & 1 \\
0 & 1 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 1 & 0 \\
1 & 0 & 0 & 1 & 0 & 0 & 1
\end{bmatrix}
=
\begin{bmatrix}
1 & 0 & 0 & 1 & 0 & 0 & 1 \\
0 & 1 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 1 & 0 \\
1 & 0 & 0 & 1 & 0 & 0 & 1 \\
0 & 1 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 1 & 0 \\
1 & 0 & 0 & 1 & 0 & 0 & 1
\end{bmatrix}$$

$M_R \otimes M_R = M_R$

$\forall i \forall j$, if $(n_{ij} = 1)$ then $(m_{ij} = 1)$. Thus R is transitive.

Subject: No: Date:

⑤ $R = \{(1,3),(2,6),(3,9),(4,12)\}$

a. not reflexive, diagonal not all 1's

b. not symmetric $M_R \neq M_R^T$

c. not transitive $M_R \times M_R \neq M_R$

⑥ $$RS = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

$$SR = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

Question 2

⑦ All functions are relations, but not all relations are functions. A function of multiple inputs can have a same output but a same input cannot have multiple outputs. A relation can have multiple inputs to multiple outputs.

⑧ i) Yes, one-to-one functions

ii) Yes, Many to one

iii) No, One To Many

iv) Yes, Many to one

v) No, one to many

⑨ $R = \{(1,6),(2,7),(3,8),(4,9),(5,10)\}$

Domain $= \{1,2,3,4,5\}$

Range $= \{6,7,8,9,10\}$

⑩(v) $f(x) = 1-2x$ The function is bijective, it is both one-to-one function, onto

e-to-one function $f(x_1) = f(x_2)$

$1-2x_1 = 1-2x_2$

$1-1 = 2x_1 - 2x_2$

$0 = 2x_1 - 2x_2$

$0 = x_1 - x_2$

$x_1 = x_2$ proven

onto function:

for every y in codomain, exists on x in domain $f(x) = y$

$y = 1-2x$

$2x = 1-y$

$x = \frac{1-y}{2}$ proven

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vi) $f(x)=5x^2-1$

$f(x_1)=f(x_2)$

$5(x_1)^2-1=5(x_2)^2-1$

$5(x_1)^2=5(x_2)^2$

$(x_1)^2=(x_2)^2$

$x_1=\pm x_2$

$y=5x^2-1$

$x^2=(y+1)/5$

$x=\pm\sqrt{\frac{y+1}{5}}$

Considering positive square root, have valid solution for every y in R

thus, function is not one-to-one function, thus it is onto function.

not bijective since not one-to-one function

∴ not bijective, not one-to-one function, onto function

vii) $f(x)=x^4$

$f(x_1)=f(x_2)$

$(x_1)^4=(x_2)^4$

$x_1=\pm x_2$

$y=x^4$

$x=\pm\sqrt[4]{y}$

valid solution for every y in R

thus, onto function

not one-to-one function, not bijective, onto function

viii) $f(x_1)=f(x_2)$

$\frac{x_1-2}{x_1-3}=\frac{x_2-2}{x_2-3}$

$(x_1-2)(x_2-3)=(x_1-3)(x_2-2)$

$x_1x_2-3x_1-2x_2+6=x_1x_2-2x_1-3x_2+6$

$-3x_1-2x_2=-2x_1-3x_2$

$x_1=x_2$ proven

function is one-to-one function

$y=\frac{x-2}{x-3}$

$xy-3y=x-2$

$xy-x=3y-2$

$x=\frac{3y-2}{(y-1)}$

$y-1\neq 0$

$y\neq 1$

when $y=1$, no value in x be assigned, not cover all value in codomain, not onto function.

∴ Not bijection because not onto function, one-to-one function, not onto function

11c ix) $f(g(x))=3(x^2-1)-1$

$=3x^2-3-1$

$=3x^2-4$

$fg(0)=3(0)^2-4$

$=-4$

$fg(1)=3(1)^2-4$

$=-1$

$fg(2)=3(2)^2-4$

$=8$

$fg(3)=3(3)^2-4$

$=23$

(x) $fg(x)=(5x-6)^2$

$=25x^2-60x+36$

$fg(0)=25(0)^2-60(0)+36$

$=36$

$fg(1)=25(1)^2-60(1)+36$

$=1$

$fg(2)=25(2)^2-60(2)+36$

$=16$

$fg(3)=25(3)^2-60(3)+36$

$=81$

(xi) $fg(x)=x^3+1-1$

$=x^3$

$f(g(0))=0^3=0$

$f(g(1))=1^3=1$

$f(g(2))=2^3=8$

$f(g(3))=3^3=27$

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Question 3 Subject:

No:
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12 (xii) $a_n = 6a_{n-1} - 9a_{n-2}$

$a_n = r^n$

$(r^n = 6r^{n-1} - 9r^{n-2}) \div r^{n-2}$

$r^2 - 6r - 9$

$r^2 - 6r + 9 = 0$

$(r-3)(r-3) = 0$

$r = 3$

$a_n = (x + yn)3^n$

$a_{n=1}$

$1 = (x + y(0))3^0$

$x = 1$

$a_1 = 6$

$6 = (x + y(1))3^1$

$6 = 3x + 3y$

$x + y = 2$

$1 + y = 2$

$y = 1$

$\therefore a_n = (1+n)3^n$

x(iii) $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$, initial conditions $a_0 = 2, a_1 = 5, a_3 = 15$

$a_n = r^n$

$(r^n = 6r^{n-1} - 11r^{n-2} + 6r^{n-3}) \div r^{n-3}$

$r^3 = 6r^2 - 11r + 6$

$r^3 - 6r^2 + 11r - 6 = 0$

$(r-1)(r^2 - 5r + 6) = 0$

$(r-1)(r-2)(r-3) = 0$

$r = 1, r = 2, r = 3$

$a_n = a + b(2^n) + c(3^n)$

$a_0 = 2,\ 2 = a + b(2^0) + c(3^0)$

$a + b + c = 2$ — ①

$a_1 = 5,\ 5 = a + b(2^1) + c(3^1)$

$= a + 2b + 3c = 5$ — ②

$a_2 = 15,\ 15 = a + b(2^2) + c(3^2)$

$a + 4b + 9c = 15$ — ③

② − ① = $b + 2c = 3$ — ④

③ − ② $2b + 6c = 10$ — ⑤

⑤ − ④×2 $2c = 4$

$c = 2$

$b + 2(2) = 3$

$b = -1$

$a + (-1) + 2 = 2$

$a = 1$

$\therefore a_n = 1 - 2^n + 2(3^n)$

(xiv) $a_n = -3a_{n-1} - 3a_{n-2} + a_{n-3}$

$a_n = r^n$

$(r^n + 3r^{n-1} + 3r^{n+2} - r^{n-3} = 0) \div r^{n-3}$

$r^3 + 3r^2 + 3r - 1 = 0$

$(r+1)^3 - 2 = 0$

$r + 1 = \sqrt[3]{2}$

$r = \sqrt[3]{2} - 1$

$a_n = (x + 4n)(\sqrt[3]{2} - 1)$

$a_0 = 1\quad 1 = (x + 4(n))(\sqrt[3]{2} - 1)^1$

$1 = x$

$a_1 = -2$

$-2 = (x + 4(1))(\sqrt[3]{2} - 1)^1$

$-2 = (\sqrt[3]{2} - 1)(x) + (\sqrt[3]{2} - 1)y$

$-2 = (\sqrt[3]{2} - 1) + (\sqrt[3]{2} - 1)y$

$\dfrac{-1 - \sqrt[3]{2}}{\sqrt[3]{2} - 1} = y$

13 $a_2 = 5k - 3$

$a_3 = 5(5k-3) - 3$

$= 25k - 18$

$a_4 = 5(25k - 18) - 3$

$= 125k - 93$

(ii) $7 = 125k - 93$

$125k = 100$

$k = 0.8$

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